

Department of Computer Science and Engineering (CSE)
BRAC University

Lecture 12

CSE250 - Circuits and Electronics

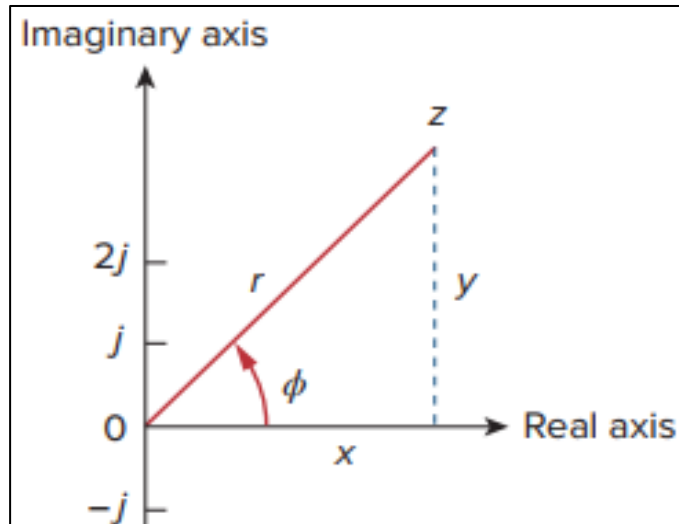
AC FUNDAMENTALS, AC CIRCUITS, AND AC POWER



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Complex Number

- A complex number z can be written as,
- $z = x + jy$ Rectangular form
- $z = r\angle\phi$ Polar form $j = \sqrt{-1} = -\frac{1}{j}$
- $z = re^{j\phi}$ Exponential form
- The relationship between the rectangular and the polar form can be written from the figure as,



- $r = \sqrt{x^2 + y^2}$
- $\phi = \tan^{-1} \frac{y}{x}$
- $x = r \cos \phi$
- $y = r \sin \phi$

- The complex exponential can be expanded by a Taylor's series expansion as,

$$e^{j\phi} = 1 + j\phi + \frac{(j\phi)^2}{2!} + \frac{(j\phi)^3}{3!} + \frac{(j\phi)^4}{4!} + \dots$$

- Separating the real and imaginary parts,

$$\Rightarrow e^{j\phi} = \left(1 - \frac{\phi^2}{2!} + \frac{\phi^4}{4!} - \dots\right) + j\left(\phi - \frac{\phi^3}{3!} + \frac{\phi^5}{5!} - \dots\right)$$

- The real and imaginary parts are the Taylor's series expansion of a cosine and a sine respectively.

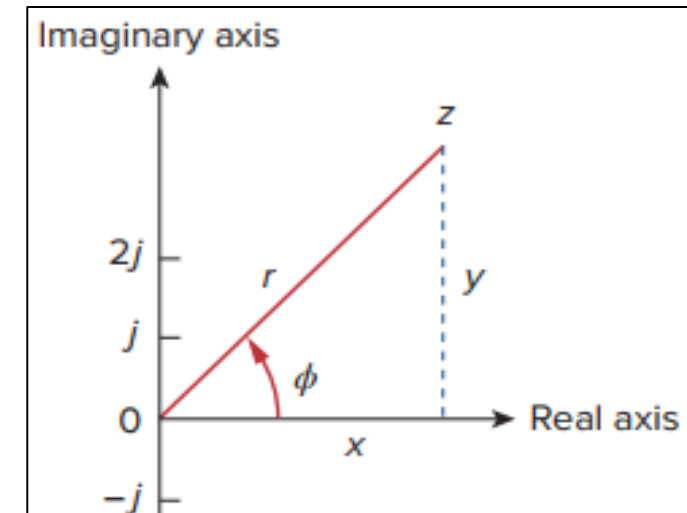
$$\Rightarrow e^{j\phi} = \cos \phi + j \sin \phi$$

$$\Rightarrow re^{j\phi} = r \cos \phi + jr \sin \phi$$

$$\text{So, } \mathbf{z = re^{j\phi} = x + jy}$$

Operations of Complex Numbers

- Rectangular form $z = x + jy$
- Polar form $z = r \angle \phi$ where, $j = \sqrt{-1} = -\frac{1}{j}$
- Exponential form $z = re^{j\phi}$
- Let, $z_1 = x_1 + jy_1 = r_1 \angle \phi_1 = r_1 e^{j\phi_1}$ and $z_2 = x_2 + jy_2 = r_2 \angle \phi_2 = r_2 e^{j\phi_2}$
- Addition: $z_1 + z_2 = (x_1 + x_2) + j(y_1 + y_2)$
- Subtraction: $z_1 - z_2 = (x_1 - x_2) + j(y_1 - y_2)$
- Multiplication: $z_1 z_2 = r_1 r_2 \angle (\phi_1 + \phi_2)$
- Division: $z_1 / z_2 = \frac{r_1}{r_2} \angle (\phi_1 - \phi_2)$
- Reciprocal: $\frac{1}{z} = \frac{1}{r} \angle -\phi$
- Square root: $\sqrt{z} = \sqrt{r} \angle \frac{\phi}{2}$
- Complex Conjugate: $z^* = x - jy = r \angle -\phi = re^{-j\phi}$



Problem 1

- Determine the polar and rectangular form of the following quantities:

$$I. \frac{(25\angle 36.9^\circ)(80\angle -53.1^\circ)}{(4+j8)+(6-j8)}$$

$$II. 5\angle 81.87^\circ \left(4 - j3 + \frac{3\sqrt{2}\angle -45^\circ}{7-j1} \right)$$

$$III. [5^{-1} + 10^{-1} + (j24)^{-1} + (1\angle -90^\circ)^{-1}]^{-1}$$

$$IV. \frac{[5^{-1} + (j12)^{-1} + 10^{-1}]^{-1}}{[5^{-1} + (j12)^{-1} + 10^{-1}]^{-1} + (-j2)} \times 10\angle 60^\circ$$

$$\text{Ans: } I. 200\angle -16.2^\circ = 192.0587 - j55.7982$$

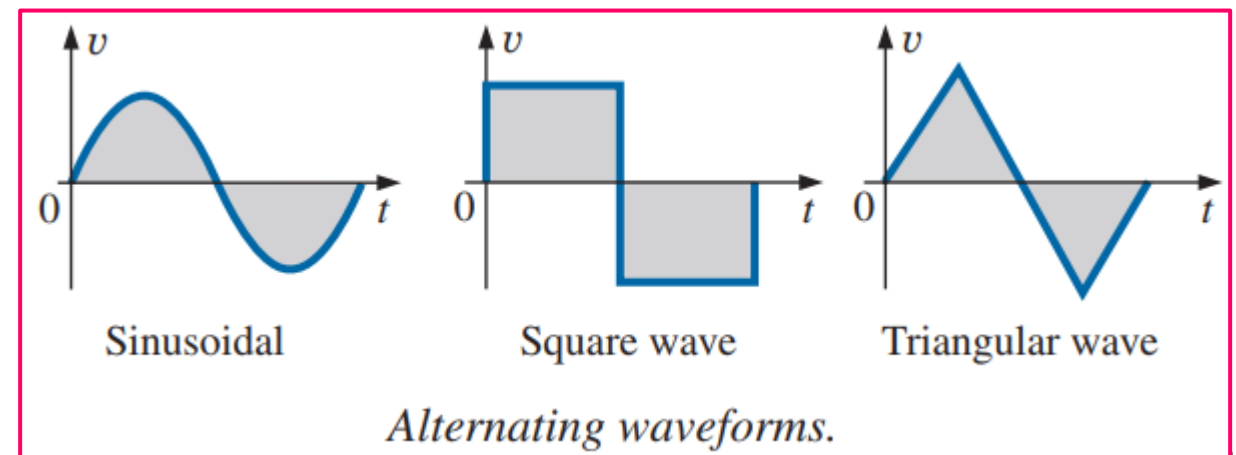
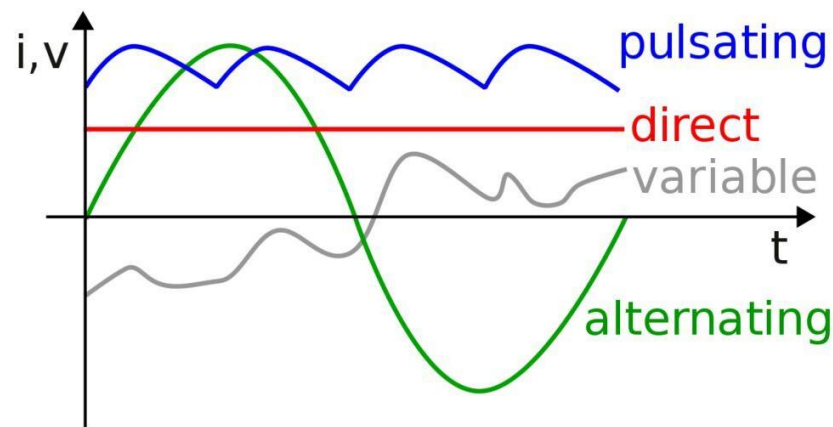
$$II. 76.2520 + j44.2506 = 88.162\angle 30.127^\circ$$

$$III. 1\angle -72.62^\circ = -0.297 + j0.95$$

$$IV. 9.738\angle 95.75^\circ = -0.976 + j9.689$$

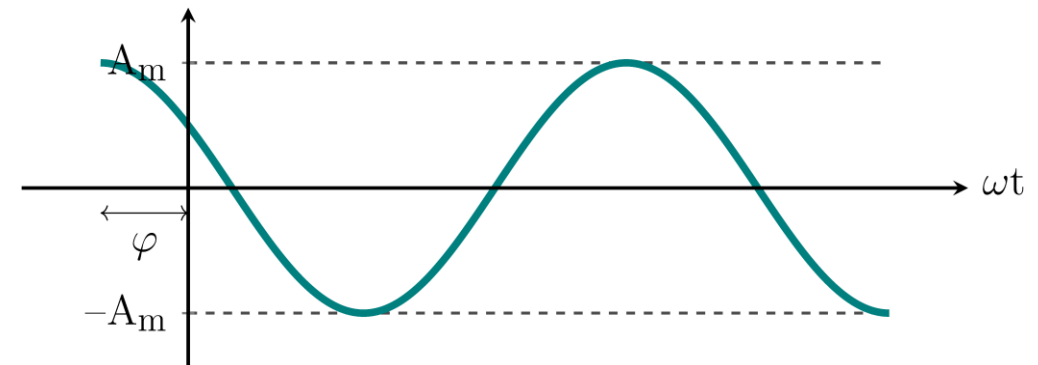
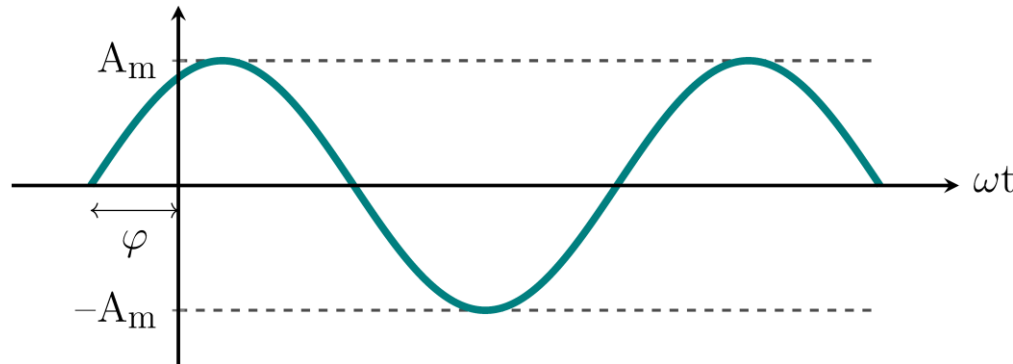
Alternating Current or Voltage

- *Alternating current (AC)* is a flow of electric charge that **periodically reverses** its direction, in contrast to direct current (*DC*) which only flows in a single direction. It starts, say, from zero, grows to a maximum, decreases to zero, reverses, reaches a maximum in the opposite direction, returns again to the original value, and repeats this cycle indefinitely.
- The interval of time between the attainment of a definite value on two successive cycles is called the *period*, the number of cycles or periods per second is the *frequency*, and the maximum value in either direction is the *amplitude* of the alternating current.



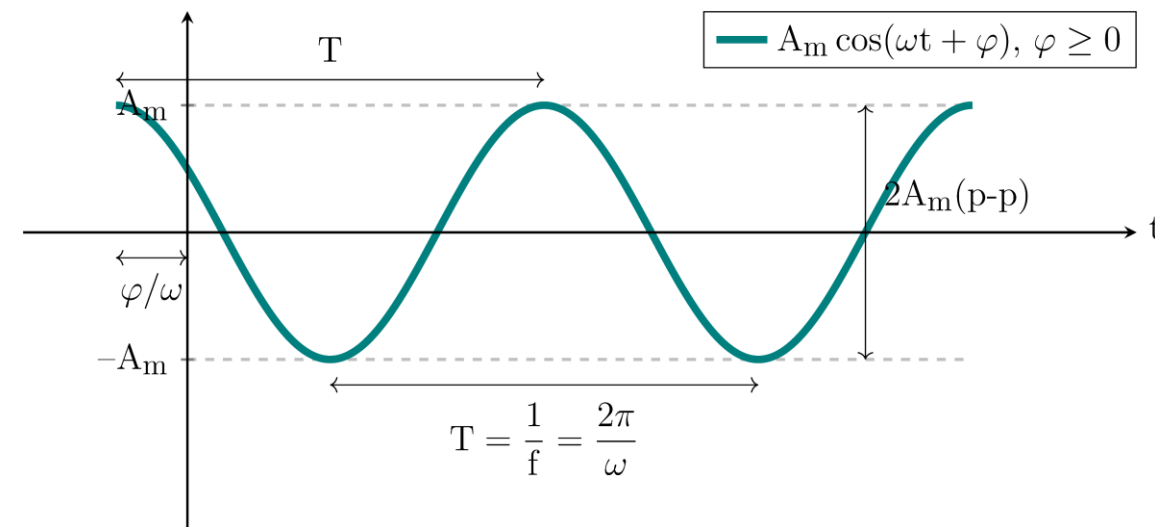
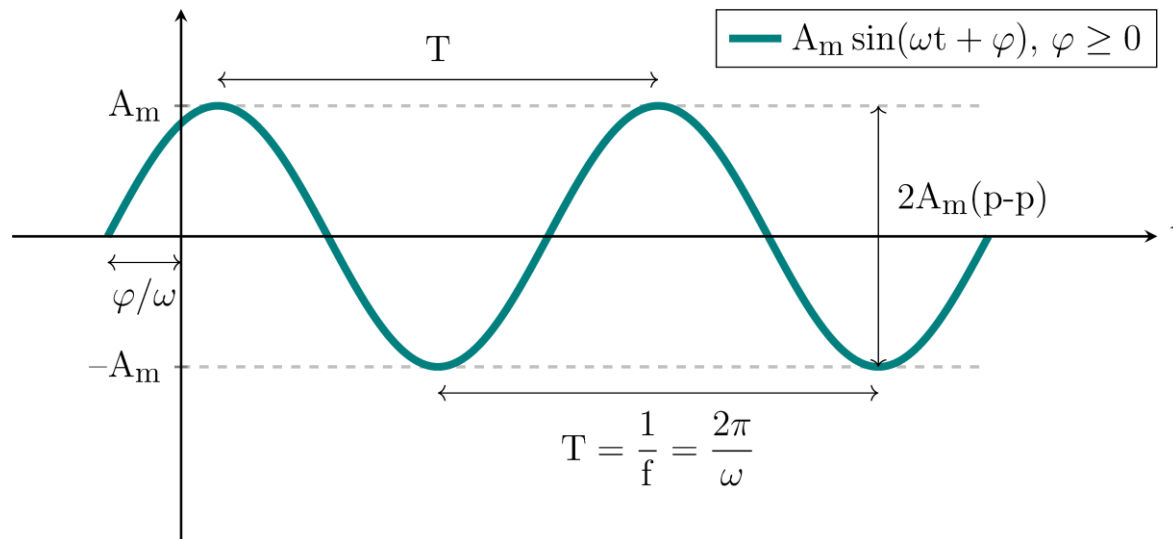
Sinusoid

- Among different types of ac waveforms, the pattern of particular interest is the sinusoidal because (i) it is the voltage generated by utilities throughout the world and supplied to homes, factories, laboratories, and so on and (ii) any periodic signal can be represented as a series of summation of sine and cosine.
- A *sinusoid* is a signal that has the form of the sine or cosine function. A sinusoidal current is usually referred to as *alternating current (ac)*. Circuits driven by sinusoidal current sources or voltage sources are called *ac circuits*.
- The basic mathematical format for the sinusoidal waveform is $A_m \cos(\omega t + \phi)$ or $A_m \sin(\omega t + \phi)$



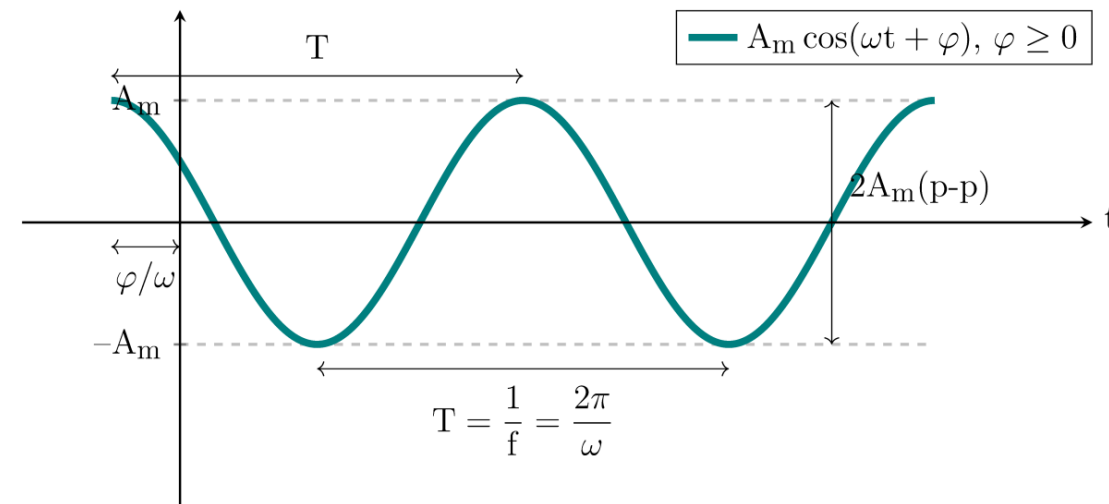
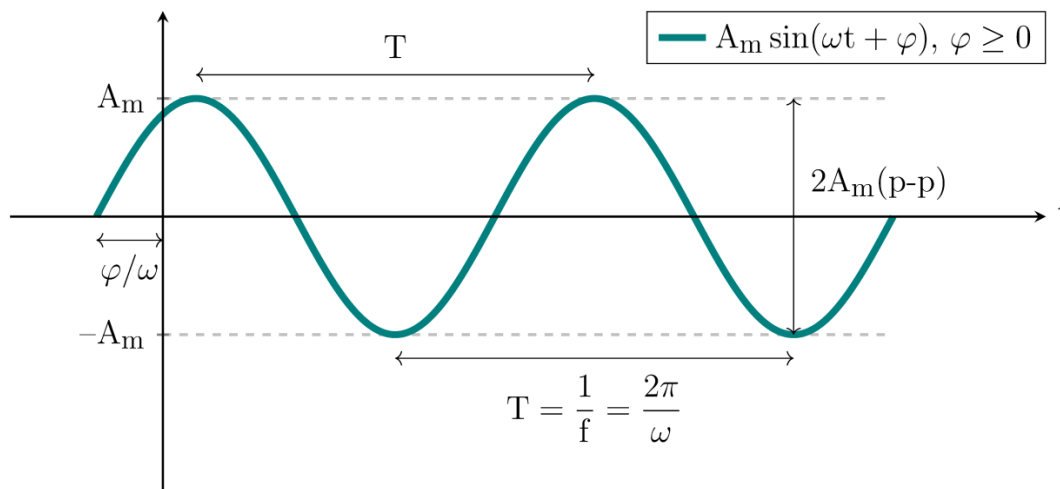
Sinusoid: basic terms

- It is necessary to define a few basic terms for a sinusoid of the form $A_m \cos(\omega t + \phi)$ or $A_m \sin(\omega t + \phi)$. These terms, however, can be applied to any alternating waveform.
- Waveform:** The path traced by a quantity plotted as a function of some variable, such as time, position, degrees, radians, temperature, and so on.
- Periodic Waveform:** A waveform that continually repeats itself after the same time interval. The waveforms shown are periodic waveforms.
- Instantaneous Value:** The magnitude of a waveform at any instant of time.



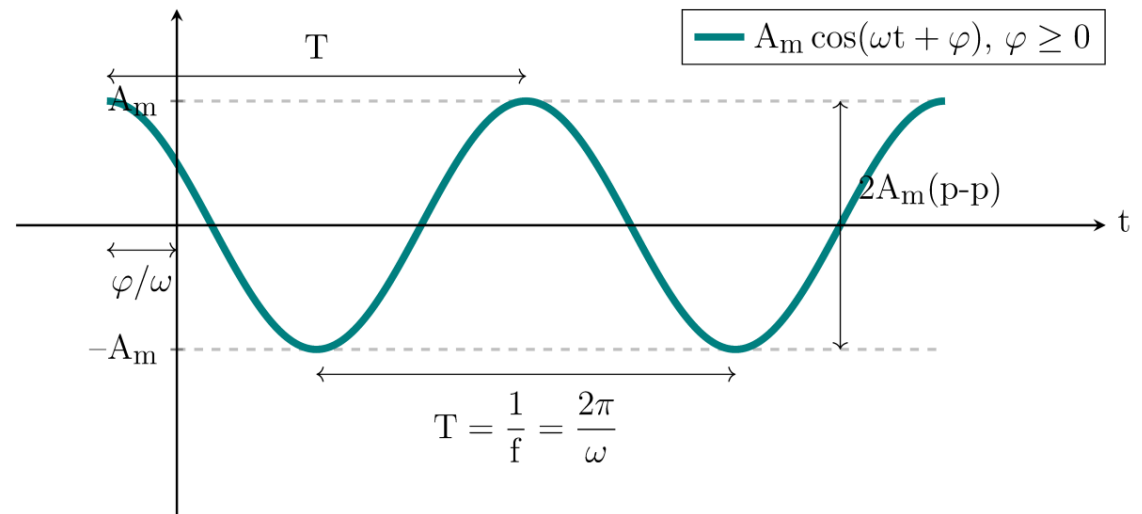
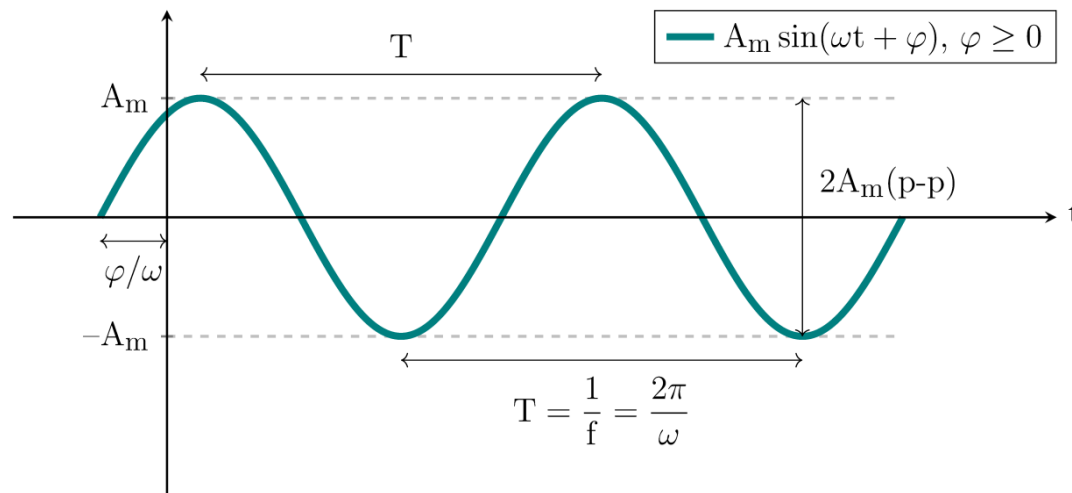
Sinusoid: basic terms (2)

- **Peak amplitude or Amplitude:** The maximum value (A_m) of a waveform as measured from its average, or mean, value.
- **Peak value or Magnitude:** The maximum instantaneous value of a function as measured from the zero volt level. For the waveforms shown, the peak amplitude and peak value are the same (A_m) since the average value of the function is zero volts.
- **Peak-to-peak value:** Denoted by $p - p$, the full voltage between positive and negative peaks of the waveform, that is, the sum of the magnitude of the positive and negative peaks.
- **Period (T):** The time interval of a periodic waveform after which the waveform repeats.

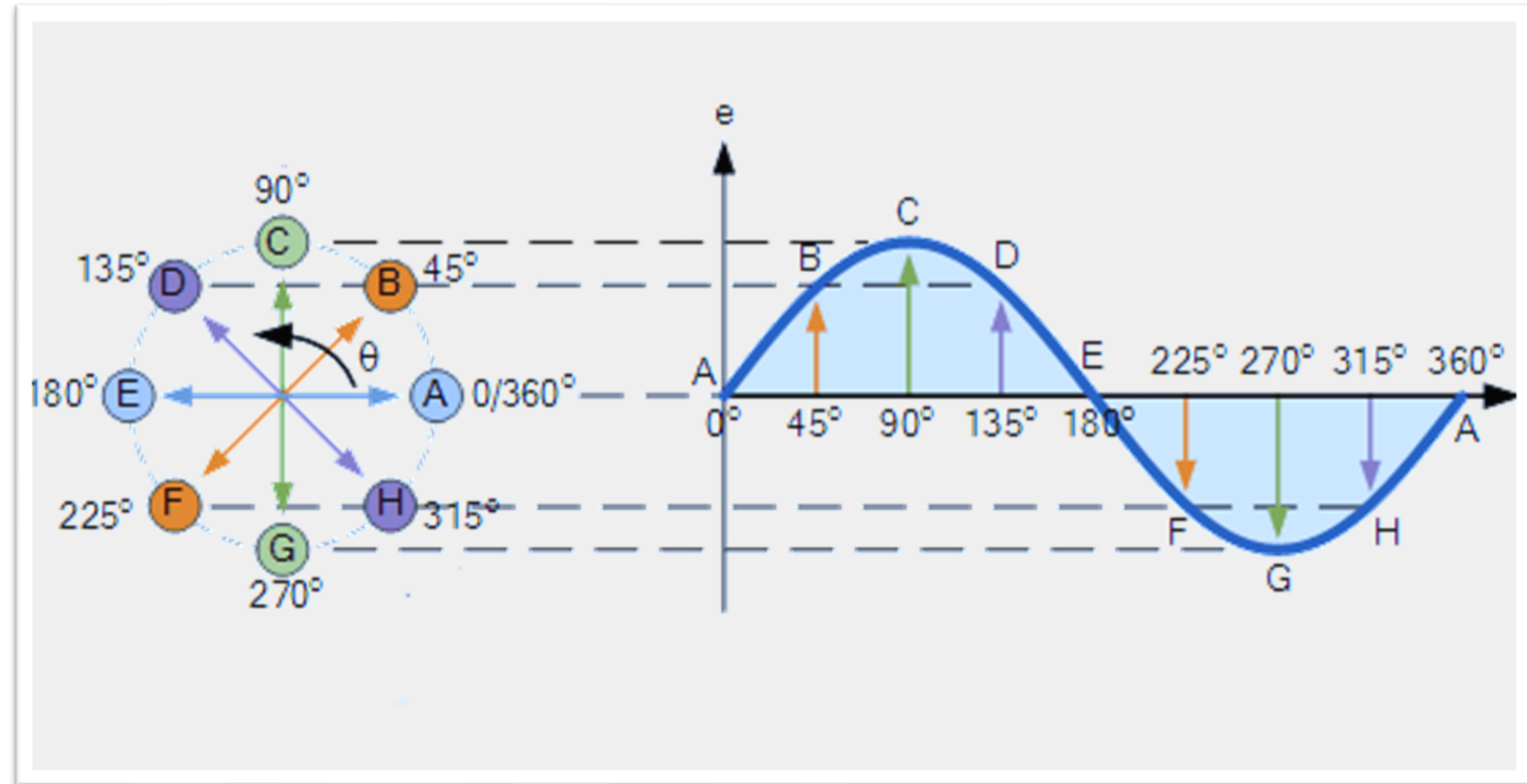
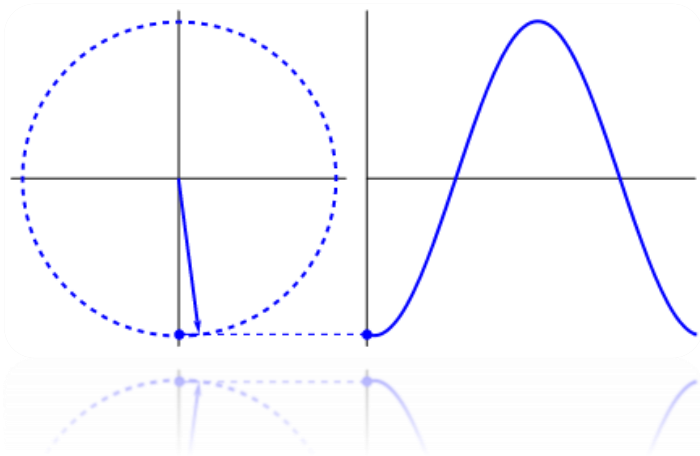


Sinusoid: basic terms (3)

- **Cycle:** The portion of a waveform contained in one period of time.
- **Natural Frequency (f):** The number of cycles that occur in 1 s. The unit of measure for frequency is the hertz (Hz), where 1 hertz (Hz) = 1 cycle per second (cps).
- **Angular frequency (ω):** Measure of angular displacement per unit of time. The unit of measure is radian per second or rads^{-1} .
- **Instantaneous phase ($\omega t + \varphi$):** Phase of a sine or cosine at any instant of time.
- **Initial phase (φ):** Phase of a sinusoid at $t = 0$.

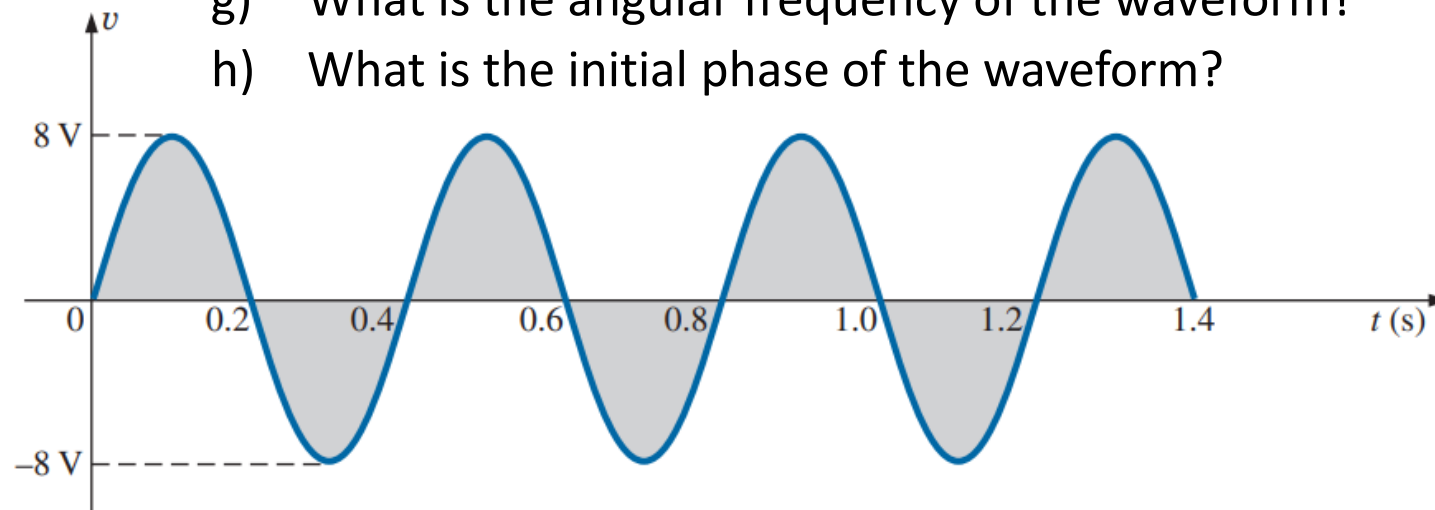


When an object moves in a circle, we can project its position onto a linear axis (like the x-axis or y-axis) to see how its position changes over time. The result of this projection is a sinusoidal wave.



Problem 2

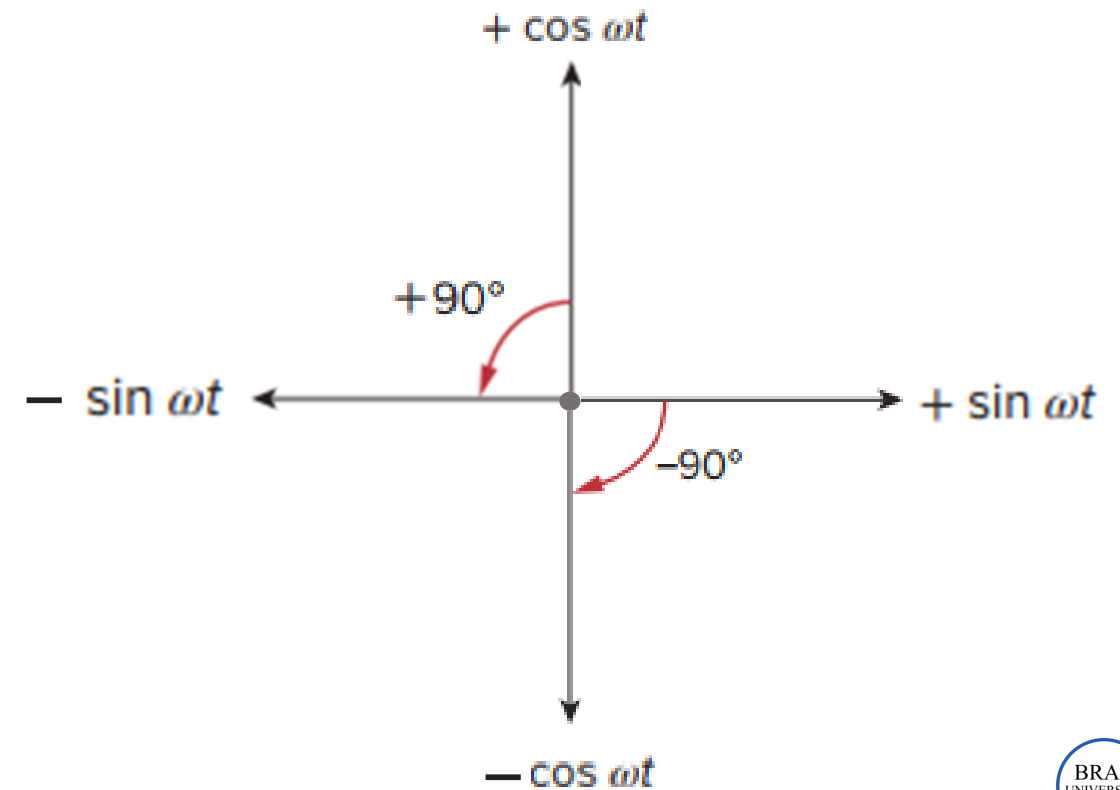
- For the sinusoidal waveform shown below,
 - a) What is the peak value?
 - b) What is the instantaneous value at 0.3s and 0.6s?
 - c) What is the peak-to-peak value of the waveform?
 - d) What is the period of the waveform?
 - e) How many cycles are shown?
 - f) What is the natural frequency of the waveform?
 - g) What is the angular frequency of the waveform?
 - h) What is the initial phase of the waveform?



Ans: Try yourself

Sine-Cosine Conversion

- A sinusoid can be expressed in either sine or cosine form. When comparing two sinusoids, it is expedient to express both as either sine or cosine with positive amplitudes.
- The following trigonometric identities can be used to convert from sine to cosine or vice versa.
- $\sin(\omega t \pm 180^\circ) = -\sin \omega t$
- $\cos(\omega t \pm 180^\circ) = -\cos \omega t$
- $\sin(\omega t \pm 90^\circ) = \pm \cos \omega t$
- $\cos(\omega t \pm 90^\circ) = \mp \sin \omega t$



Problem 3

- Calculate amplitude, **initial phase** ($-180^\circ \leq \phi \leq 180^\circ$), angular frequency, period, and frequency for the following sinusoids. What are the values of V_s and I_s at $t = 20 \text{ ms}$?

[Both sin and cosine forms can be employed]

I. $V_s = 45 \cos(5\pi t + 36^\circ) \text{ (V)}$

II. $I_s = 15 \cos(25\pi t + 25^\circ) \text{ (A)}$

III. $I_s = -20 \cos(314t - 30^\circ) \text{ (A)}$

IV. $V_s = -4 \sin(628t + 55^\circ) \text{ (V)}$

Ans: I. 45 V; **36°** or **126°**; $5\pi \text{ rad/s}$; 0.4 s; 2.5 Hz; **26.45 V**

II. 15 A; **25°** or **115°**; $25\pi \text{ rad/s}$; 80 ms; 12.5 Hz; **- 6.34 A**

III. 20 A; **150°** or **60°**; 314 rad/s ; 20 ms; 50 Hz; **- 17.29 A**

IV. 4 V; **- 125°** or **60°**; 628 rad/s ; 10 ms; 100 Hz; **- 3.26 V**

Initial phase from, **Cos equation** and **Sin Equation**

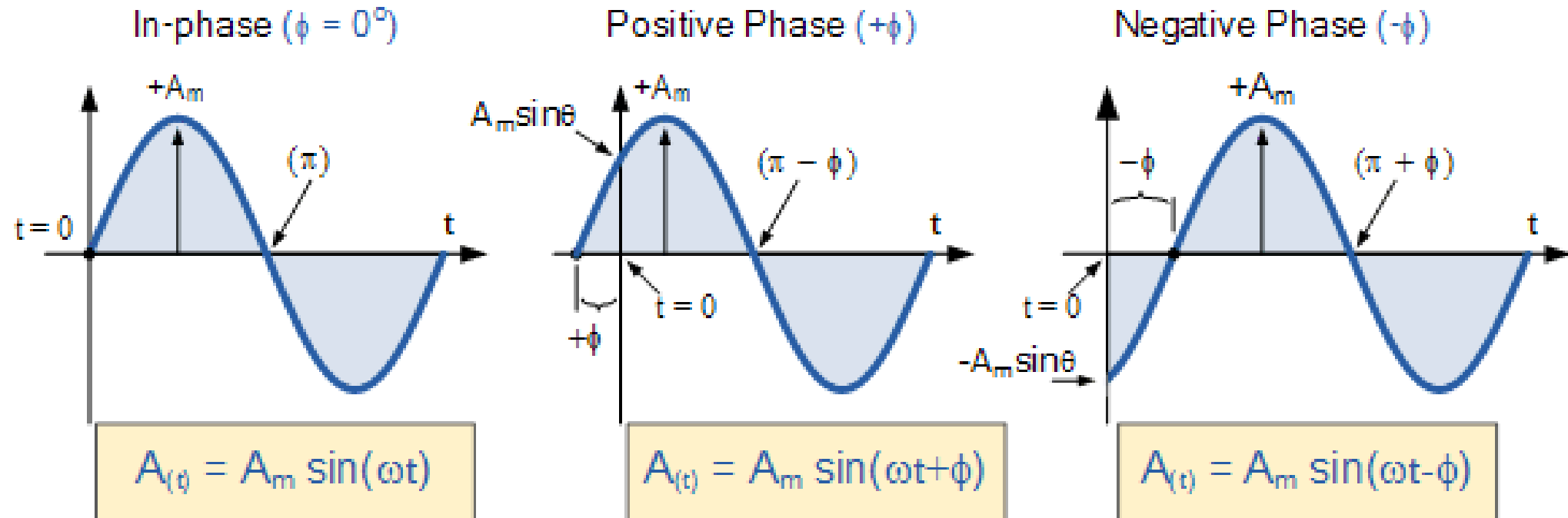
MUST REMEMBER, The terms inside the cosine or sine function must be of same unit. If not, you have make them same unit before doing calculations. For example, for the first problem. If we want to calculate V_s at $t = 20 \text{ ms}$. The equation will be,

$$V_s(20 \times 10^{-3}) = 45 \cos\left(5 \times \pi \times \frac{180}{\pi} \times 20 \times 10^{-3} + 36^\circ\right) \text{ (V)}$$

See here, we multiplied π with $\frac{180^\circ}{\pi}$, Why? Because the initial phase is given in degree. And to sum/subtract, we need to have units. So we converted the first term in degree as well.

From a circle,

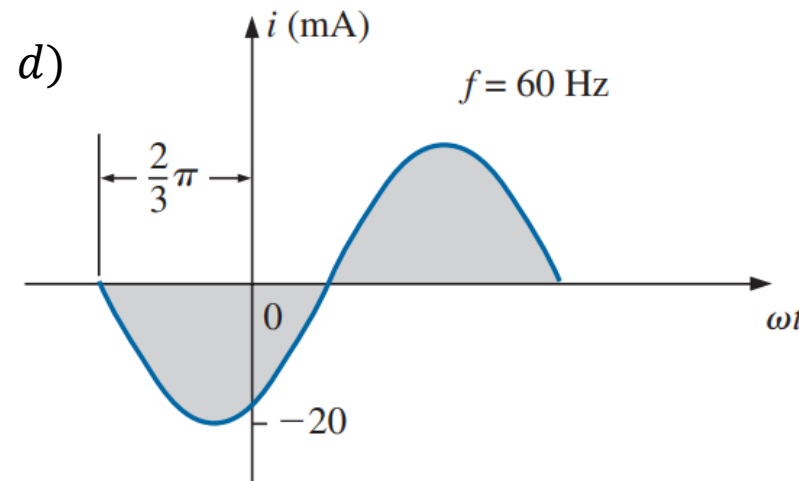
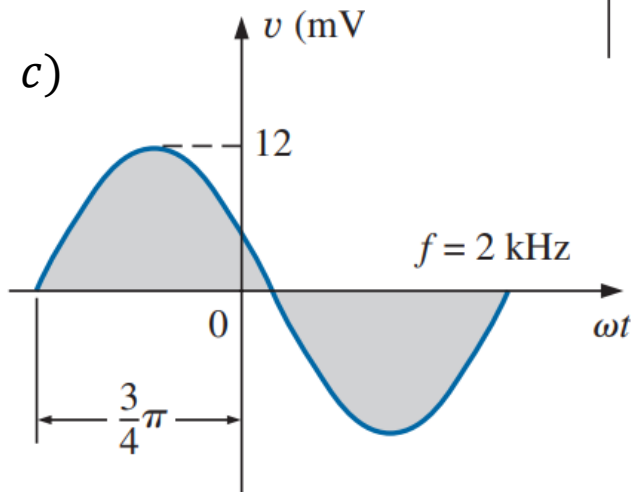
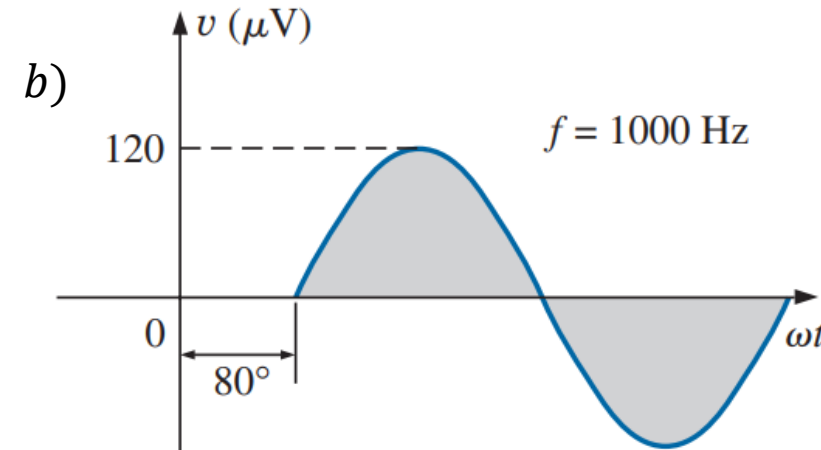
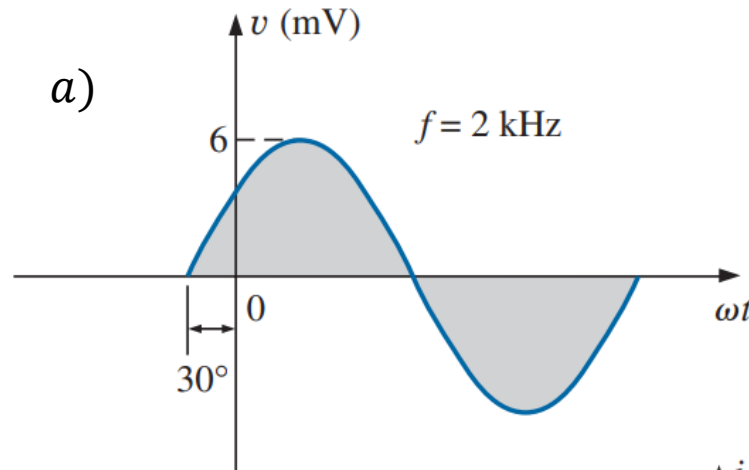
$$2\pi = 360^\circ$$



Problem 4

Inverted or non inverted, does not matter. For initial phase, just look at where the y axis was intersected

- Write analytical expressions for the waveforms with the initial phase in degrees.



Ans:

a) $6 \times 10^{-3} \sin(2\pi 2000t + 30^\circ) \text{ V}$

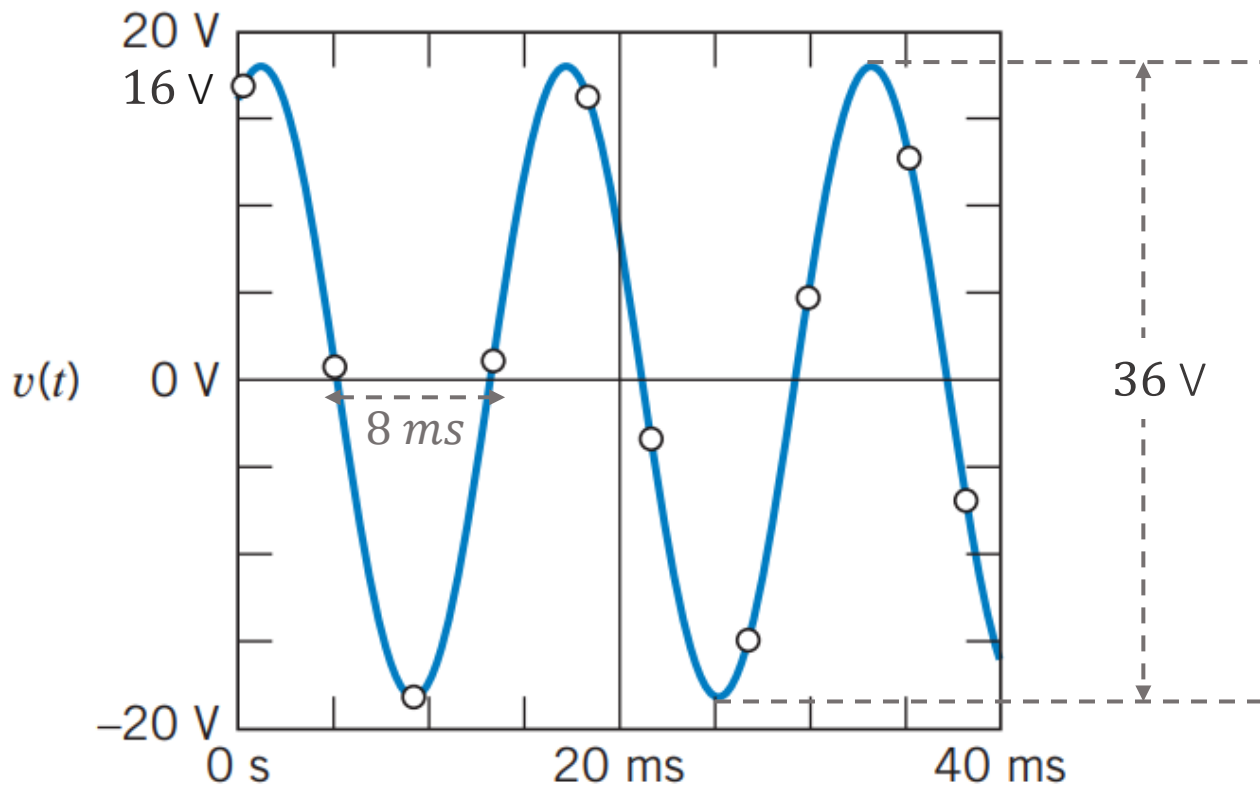
b) $120 \times 10^{-6} \sin(2\pi 1000t - 80^\circ) \text{ V}$

c) $12 \times 10^{-3} \sin(2\pi 2000t + 135^\circ) \text{ V}$

d) $20 \times 10^{-3} \sin(2\pi 60t - 60^\circ) \text{ A}$

Problem 5

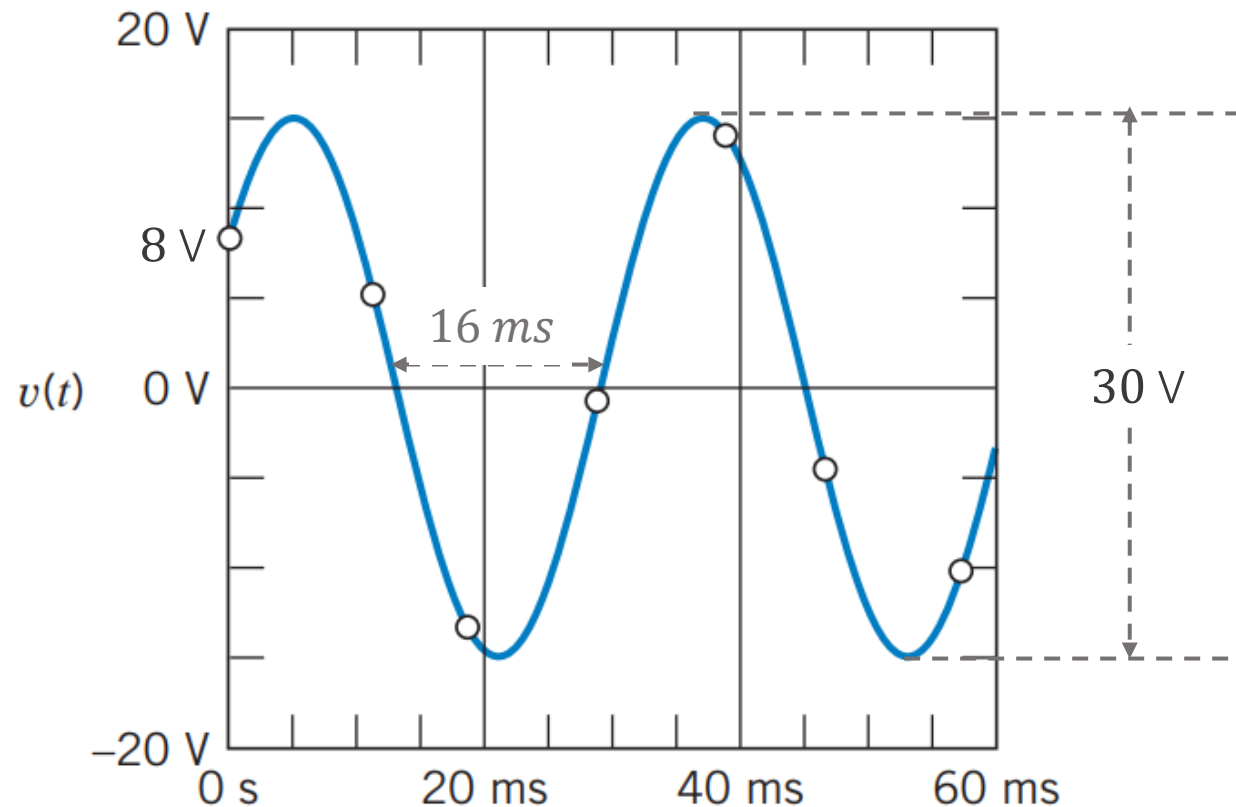
- The following figure shows a sinusoidal voltage $v(t)$, plotted as a function of time t . Represent $v(t)$ by a function of the form $A_m \cos(\omega t + \varphi)$ with φ in degrees.



Ans: $v(t) = 18 \cos(125\pi t - 27^\circ) \text{ V}$

Problem 6

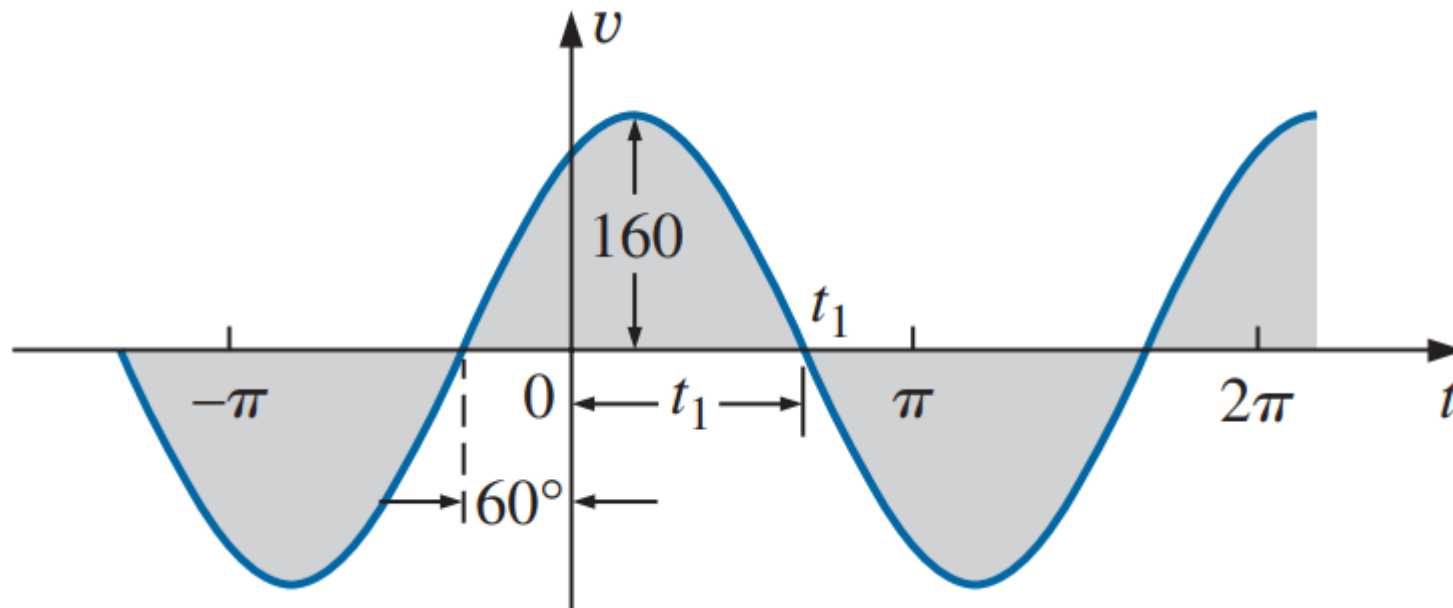
- The following figure shows a sinusoidal voltage $v(t)$, plotted as a function of time t . Represent $v(t)$ by a function of the form $A_m \sin(\omega t + \varphi)$ with φ in degrees.



Ans: $v(t) = 15 \sin(62.5\pi t + 32.23^\circ) \text{ V}$

Problem 7

- A sinusoidal voltage $v(t) = 160 \sin(2\pi 1000t + 60^\circ)$ is plotted as a function of time t below. Determine the time t_1 when the waveform crosses the axis.



Ans: $t_1 = 0.333 \text{ ms}$

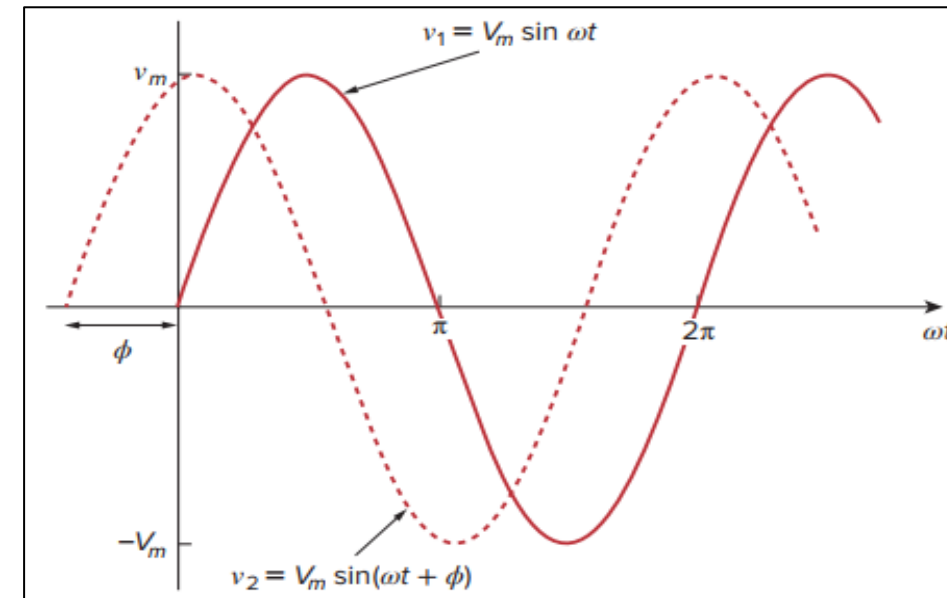
Leading and Lagging Sinusoids

- The equation for a sinusoid in more general form,

$$v(t) = V_m \sin(\omega t + \phi)$$

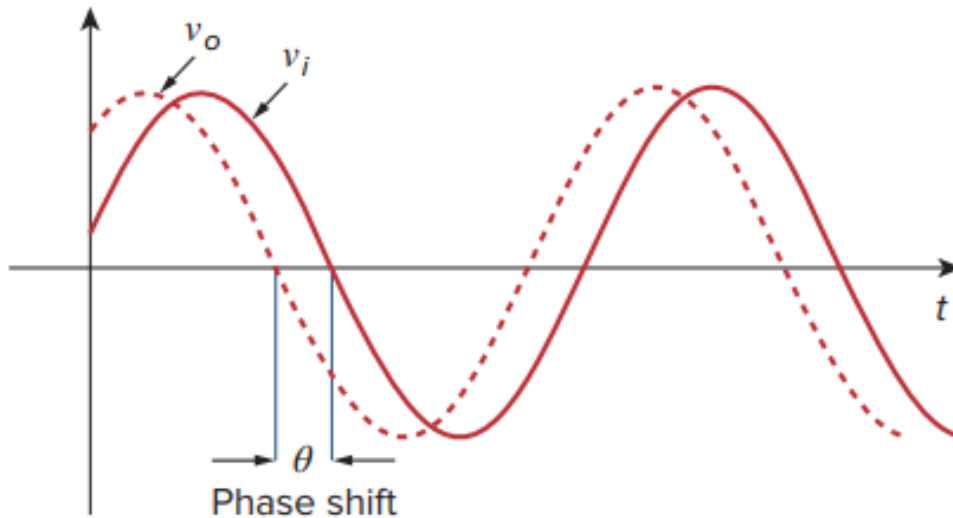
where $(\omega t + \phi)$ is the argument and ϕ is the initial phase. Both argument and phase can be in radians or degrees.

- Let's examine two sinusoids $v_1(t) = V_m \sin \omega t$ and $v_2(t) = V_m \sin(\omega t + \phi)$
- The starting point of v_2 occurs first in time. **Or v_2 passes the zero-crossing line first** if compared between two same phase points of v_1 and v_2 .
- Therefore, we say that v_2 leads v_1 by ϕ or that v_1 lags v_2 by ϕ .
- If $\phi \neq 0$, v_1 and v_2 are out of phase.
- If $\phi = 0$, v_1 and v_2 are in phase

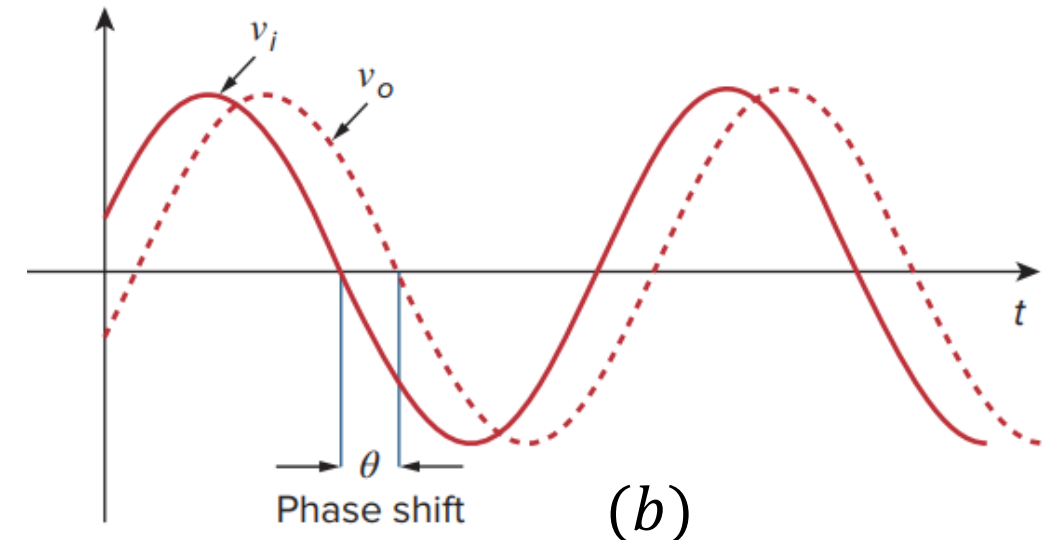


Problem 8

- Determine for each of the plots, which one is leading/lagging.

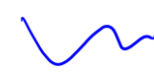


(a)



(b)

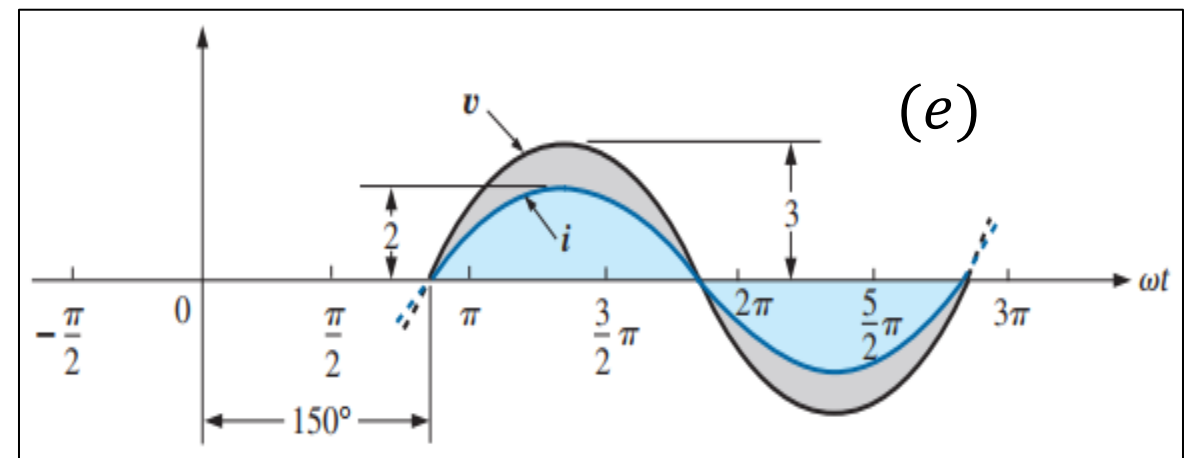
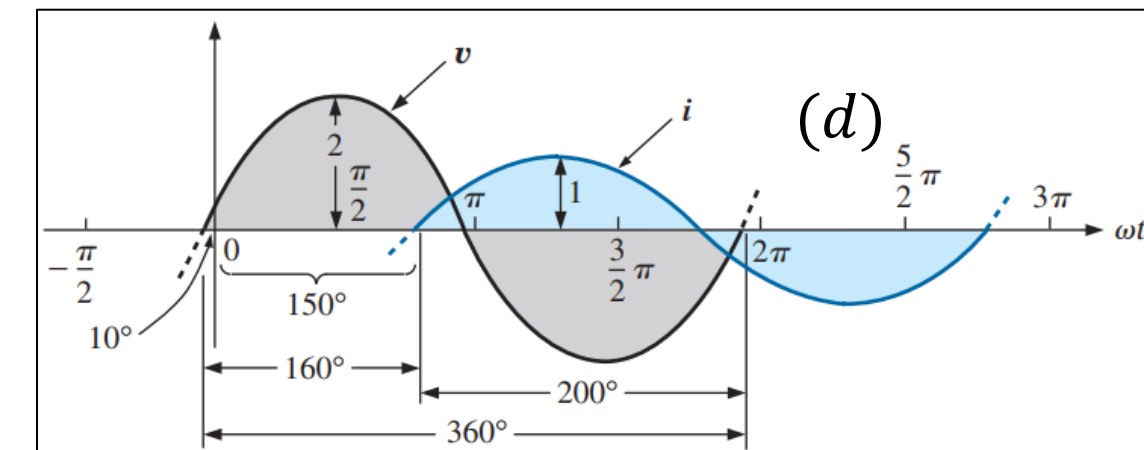
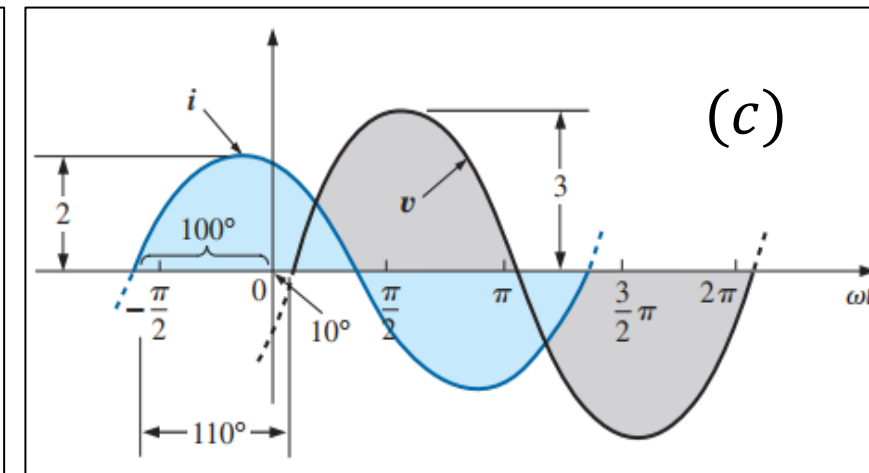
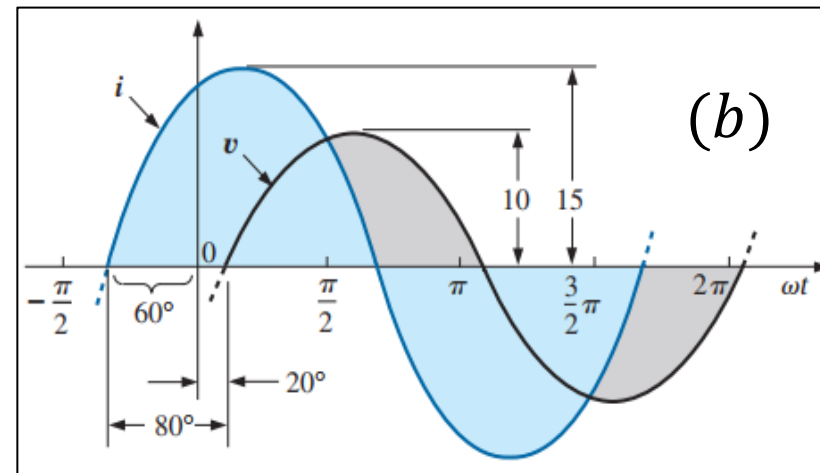
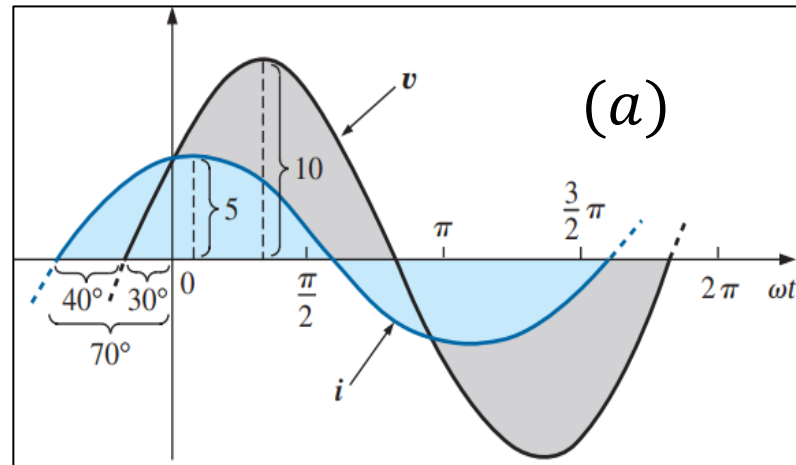
Ans: (a) v_o leading; (b) v_i leading



Problem 9

Ans: (a) i leads v by 40° ; (b) i leads v by 80° ; (c) i leads v by 110° ; (d) v leads i by 160° ; (e) v and i are in phase

- Determine for each of the plots, which one (v or i) is leading/lagging and by how much.



Problem 10

- For the following pairs of sinusoids, determine which one leads and by how much ($0 \leq \theta \leq 180^\circ$).

I. $v(t) = 10 \cos(4t - 60^\circ)$ & $i(t) = 4 \sin(4t + 50^\circ)$

II. $v_1(t) = 4 \cos(377t + 10^\circ)$ & $v_2(t) = -20 \cos 377t$

III. $v_1(t) = 45 \sin(\omega t + 30^\circ) V$ & $v_2(t) = 50 \cos(\omega t - 30^\circ)$

IV. $i_1(t) = -4 \sin(377t + 55^\circ)$ & $i_2(t) = 5 \cos(377t - 65^\circ)$

V. $x(t) = (13 \cos 2t + 5 \sin 2t)$ & $y(t) = 15 \cos(2t - 11.8^\circ)$

Hint: convert both the sinusoids into sine or cosine form → convert them into phasors → add them in frequency domain → convert them back in the time domain and compare

Ans: I. i leads v by 20°

II. v_2 leads v_1 by 170°

III. v_2 leads v_1 by 30°

IV. i_2 leads i_1 by 150°

V. y leads x by 9.24°

Practice

Books: Introductory Circuit Analysis (11th edition)

Chapter 13

Link: <https://drive.google.com/drive/folders/1k7JLZnhEzP2cMwrjG9ggIHJxUCUYcZXM?usp=sharing>

Example:

13.1, 13.2, 13.3, 13.4, 13.5, 13.6, 13.7

Exercise:

1, 2, 3, 25, 26, 27, 28, 34, 35